

What is the significance of 'Quantum Mechanics' in nanotechnology?

Phenomena in bulk matter and particles are explained on the basis of classical physics where mass, volume, time and energy are related through exact equations. Hence it is called as deterministic physics. The laws applicable to atomic and sub atomic particles are less certain and hence termed as probabilistic. These are governed by quantum mechanics.

Nanophase matter is made up of only a few atoms and the properties are dominated by the surface atoms. In turn, the surface of these atoms is made up of outer shell electrons. The behavior of the electrons is understood in terms of quantum mechanics. Hence, the principles of quantum mechanics, as applied to electrons, provide an important basis for interpreting nano-scale phenomena.

The following are the principles of quantum mechanics applied to electrons:

- Electrons behave as particles as well as waves and exhibit phenomena like interference.
- The energy of an electron can vary only by certain discrete values. Thus the energy levels of the electrons are quantized.
- The position and the momentum of an electron cannot be determined precisely. Higher the certainty in position, the higher the uncertainty in momentum; and vice versa.
- Wavelength of the wave associated with an electron is inversely proportional to its momentum.
- The probability of existence of an electron at a given point in 3-co-ordinate space, at a given time, is described by (the square) of the 'wave function'.

Comprehension of these quantum mechanical concepts will facilitate better understanding of the nano-scale phenomena such as quantum confinement and quantum tunneling.

Quantization in Nanomaterials

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Abstract—This paper is a review of quantization phenomena in nanomaterials. Dimensions of nanomaterial are reviewed and Quantum concept is analyzed. The influence of band structures on the physical properties of quantized structures is studied. Energy level and effective mass calculation through equations are understood. Finally quantized electronic structure of semiconductor is studied.

Keywords—nanomaterial,quantization, emiconductor,quantum ,band structure.

INTRODUCTION

The physical properties of the material significantly depend on their size when reduced to Nano dimension. In bulk system, most of the atoms are the interior atoms, so the properties of the materials are determined by them. But in the case of nanoparticles, the fraction of surface atoms cannot be neglected. The nonmaterial are the elements that have at least one spatial measurements in size range 1 to 100 nanometers. The quantum number determines a set of discrete energies which the electron can have, since the energy is directly related to the electron's momentum. Quantum confinement is responsible for the increase of energy difference between energy states and band gap. Nanomaterials are generally characterised as zero dimensional nanomaterials and one dimensional nanomaterials. Low dimensional structures having quantum confinement of one, two, and three dimensions such as

- quantum wells
- quantum wires
- Quantum dots

To predict the confined energy levels of nanostructures Schrodinger equations are used. In recent years, with the advent of MBE, MOCVD and other experimental techniques, the influence of quantization of band states on the different physical properties of nanostructured materials such as quantum wells, quantum well wires, quantum dots, inversion layers, magnetic quantization, and different field added dimensionally reduced systems. The influence of band structures on the physical properties of quantized structures is becoming increasingly important

INTRODUCTION TO NANOMATERIALS

There are different kinds of materials of different size around us. Some materials are natural e.g. silver, gold, and garnet etc. and some materials are manmade e.g. cement, CD and pigments etc. Human eye can see the materials with naked eye using visible rays 350 nm to 700 nm. Human eye resolution is 0.07mm. If we want to see the material below 0.07 mm size then we have to use different microscopes according to the size of material.

What is Nanoscale?

It can be used to refer to one billionth of any measure.

Nanometer

Nanosecond

Nanogram

Nanomole

The diameter of one human hair is about 10,000 to 80,000 nanometers. The diameter of atom is 0.1 nm to 0.2 nm. A crystal of 0.1 mm size contains approximately 1021 atoms. Nanoscale materials have size less than 100 nm. Nanoscale materials can be seen using light rays of wavelength in the range of 0.1 nm to 10 nm which can be achieved using electrons. So electron microscopy is very important in analysis of nanoscale materials.

Unusual properties are seen at nanometer size. For example- electrons which are delocalized becomes confined so the properties depending on localization of electron will change. Metal becomes non-metal due to this. Similarly reactivity will change due to change of surface to volume ratio. Disorder of dipoles increases with decrease of size which will result in transition from magnetic to nonmagnetic material.

Different size of gold particles appears in different colors which can be explained with the help of nanoscience.

Gold cluster < 1nm Orange gold

Gold nanoparticles: 3 to 30 nm Red color gold

More than 50 nm size: Crimson to blue color gold

Bulk gold film: Yellow color gold.

Different type of microscopes used in nanoscale analysis is

- 1- TEM- Transmission Electron Microscope(1931)
- 2- SEM- Scanning Electron Microscope(1937)
- 3- STM- Scanning Tunneling Microscope(1981)
- 4- AFM- Atomic Force Microscope(1986)

Nanotechnology gives new array of properties which are not possible with bulk material. Eric Drexler has written a book in 1981 with title "molecular manufacturing" and another book in 1986 with title "Engine of creations". These books have given the direction to nanoscale manufacturing. Atomic manipulation of Xe atoms is done in 1989 at IBM.

In electronics size is very important. Nanoscale miniaturization will lead to reduction of effective electron path and reduction in electron scattering which will lead to faster operation of electron devices but it will require more complicated manufacturing process and it will lead to increase in the fabrication cost.

Electronic and optical properties of materials can be controlled by controlling their size and shape.

III ENERGY LEVEL IN ATOMS

The sum of the kinetic and potential energies of an electron moving around a nucleus can only assume a set of discrete values for the simplest atom (hydrogen). This is due to wave like properties exhibited by electrons.

The electron wave must repeat itself exactly when followed around any circular path enclosing the nucleus to become stable. It leads to the conclusion that an integral number of wavelengths must fit into the path, and is termed as 'quantization condition'.

The electron's wavelength (λ) is inversely proportional to its momentum (p) as defined by De Broglie's relation of the wave-

particle duality:

$$\lambda = \frac{h}{p} \quad (2.1)$$

Where h is called Planck's constant and it has the value 6.6×10^{-34} Js. The quantization condition fixes the electron momentum to one of a set of discrete values depending on the number of wavelengths fitting the circular path. The quantum number determines a set of discrete energies which the electron can have, since the energy is directly related to the electron's momentum.

IV EFFECTIVE MASS:

The concept of mass of the carriers occupies a central position in the whole field of solid-state electronics. This mass is different than that of free electron mass and is called the effective mass of the carrier whose expression changes under different physical conditions. The effective carrier mass, in general, can be expressed as

$$\text{effective carrier mass along a particular direction } (m^*) = \frac{\text{momentum}(p) \text{ along this direction}}{\text{velocity}(v) \text{ along the same direction}} \quad \text{eqn. (2.2)}$$

From eqn. (2.1) we can write,

$$p = \frac{h}{\lambda} = \left(\frac{h}{2\pi} \right) \left(\frac{2\pi}{\lambda} \right) \quad (2.3)$$

The term $\left(\frac{h}{2\pi} \right)$ is called $\hbar$ and is called the normalized Planck's constant or the Dirac's constant and the term $\left(\frac{2\pi}{\lambda} \right)$ is known as the carrier wave vector $(\vec{k})$. Therefore the eqn. (2.3) can be expressed as,

$$\vec{p} = \hbar \vec{k} \quad (2.4)$$

The velocity as written in eqn (2.2) must be the group velocity $\left(\frac{\partial \omega}{\partial k} \right)$, where the frequency $\omega = \frac{E}{\hbar}$ in which E is the total energy of the carrier and

not at all the phase velocity. Therefore the velocity of the carrier is $\frac{\partial \omega}{\partial k}$. Thus the mass of the carrier should, in general, be written as

$$m^* = \frac{\hbar k}{\frac{\partial \omega}{\partial k}} = \frac{\hbar k}{\frac{1}{\hbar} \frac{\partial E}{\partial k}} \quad (2.5)$$

Therefore the effective mass of the carriers can be expressed as,

$$m^* = \hbar^2 k \frac{\partial k}{\partial E} \quad (2.6)$$

From eqn. (2.6), we observed that effective mass changes with the slope of the E-k curve. This E-k relation is called the dispersion relation which changes from semiconductor to semiconductor, and consequently the m^* also changes. The momentum effective mass is

$$m_f^* = \hbar^2 \left/ \frac{\partial^2 E}{\partial k^2} \right.$$

the most fundamental. Acceleration effective mass

This is also called the curvature effective mass. These two definitions yield the same result when $E \propto k^2$ i.e. E-k relation is parabolic. For any deviation from the parabolicity these two definitions of the effective mass will not converge to the same expression. The effective momentum mass of the carriers as given by eqn. (2.6) affects all the properties of semiconductors such as electronic heat capacity, diffusivity to mobility ratio, the Hall co-efficient, all types of transport co-efficient and changes with electron concentration and other externally controllable parameters.

V FERMI-DIRAC STATISTICS

The Fermi-Dirac statistics expresses the probability with which the electron will occupy the energy level E and is extremely important in the whole field of solid-state electronics.

This equation is called the functional equation and can be satisfied if and only if we substitute $f(E)$ as

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}}$$

where E_F is the fermi energy (a quantity defined later on), k_B in the Boltzman constant (8.62×10^{-5} eV/K) and T is the absolute temperature in Kelvin (K) .

The total energy E of a conduction electron is given by

$E = \text{Kinetic energy} + \text{Potential energy}$

$$= \frac{p^2}{2m_c^*} + \text{P.E. (where } m_c^* \text{ is the effective mass of the electron in the conduction band)}$$

$$= \frac{p^2}{2m_c^*} \quad (\text{assuming the energy is measured from the edge of the conduction band } E_C)$$

$$E = \frac{\hbar^2 k^2}{2m_c^*} \quad (2.8)$$

Since we have already proved that $p = \hbar k$ therefore we can write that

This indicates a parabolic dependency between the energy and wave vector as shown in the fig below and is termed as electron parabola. In this figure the horizontal line EC indicates the edge of the conduction band, the horizontal line EV is called edge of valence band, the dotted horizontal line near EC is called the donor level ND, the dotted horizontal line near EV is called the acceptor level NA. Ed is called the donor energy level, Ea is the acceptor energy level. The energy is measured from the edge of the conduction band EC in the vertical upward direction. The band gap Eg is defined as $E_g = E_C - E_V$. For positive band gap $E_C > E_V$, for negative band gap $E_C < E_V$ and for zero band gap $E_C = E_V$.

Below the line EV the heavy hole, light hole and split-off hole parabolas have been drawn, since, only three types of holes exist experimentally. The effective masses of the three types of holes are m_{hh} , m_{lh} , and m_{soh} respectively.

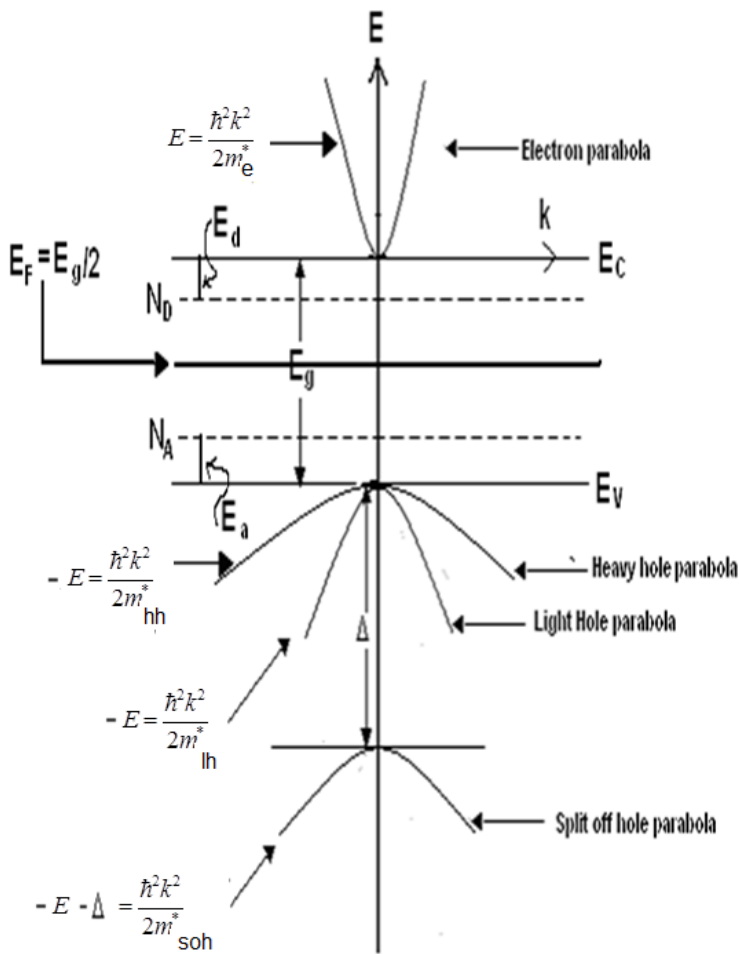


Figure 2.1 Typical model of an energy band structure of the crystalline material (Δ is called spin orbit splitting constant of valance band) Semiconductors ($E_g \leq 2.34\text{eV}$)

VI QUANTIZED ELECTRONIC STRUCTURE

Quantum confinement leads to collapse of the continuous energy bands of bulk material into discrete, atomic like energy levels. The discrete structure of energy states leads to a discrete absorption spectrum, which is in contrast to the continuous absorption spectrum of a bulk semiconductor as shown in Fig.3.1

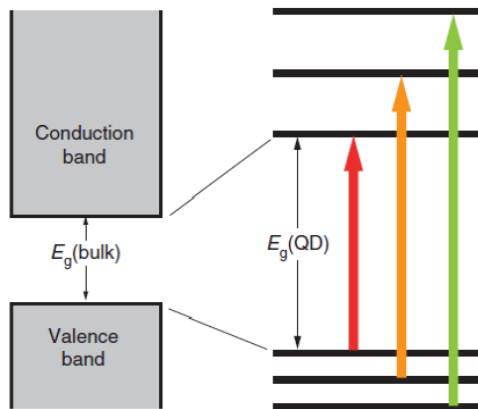


Fig. 3.1: A schematic of the discrete energy level of a semiconductor.

In semiconductors, the quantization effect that enhances the optical gap is routinely observed for clusters ranging from 1 nm to almost 10 nm. Metal particles consisting of 50 to 100 atoms with a diameter between 1 and 2 nm start to lose their metallic behavior and tend to become semiconductors.

Low dimensional structures having quantum confinement of one, two, and three dimensions (such as quantum wells, wires, and dots) have, attracted much attention in uncovering new phenomena and their interesting applications in nanotechnology.

Table 3.1 Classification of quantum confined structures.

Structure	Quantum confinement	Number of free dimension
Bulk	0	3
Quantum well/superlattices	1	2
Quantum wire	2	1
Quantum dot/Nanocrystals	3	0

In general, quantum wires, wells and dots are grown by advanced epitaxial techniques in nanocrystals produced by chemical methods or by ion implantation, or in Nano devices made by state of- the-art lithographic techniques.

In QWs, the restriction of the motion of the carriers in the two directions may be viewed as carrier confinement by two infinitely deep 1D square potential wells, leading to quantization [known as quantum size effect (QSE)] of the wave vectors along the two directions, allowing 1D electron transport representing new physical features not exhibited in bulk semiconductors.

The allowable energy levels, motion, and optical properties of the electrons are strongly affected by the quantum-mechanical effects. These artificially synthesized quantum structures find major application in high-performance transistors such as the microwave high-electron-mobility transistor (HEMT), and in high-performance solid-state lasers such as the semiconductor quantum-well laser.

Zero-dimensional materials: Nanocrystals absorb light then re-emit the light in a different colour. The size of the nanocrystal (at the Angstrom scale) determines the colour. A quantum dot is a semiconductor whose excitations are confined in all three spatial dimensions. As a result, they have properties that are between those of bulk semiconductors and those of discrete molecules.

In an unconfined (bulk) semiconductor, an electron-hole pair is typically bound within a characteristic length called the Bohr excitation radius. If the electron and hole are constrained further, then the semiconductor's properties change. This effect is a form of quantum confinement, and it is a key feature in many emerging electronic structures. Specifically, the effect describes the phenomenon results from electrons and electron holes being squeezed into a dimension that approaches a critical quantum measurement. For 0-D nanomaterials, where all the dimensions are at the nanoscale, an electron is confined in 3-D space. Quantum confinement is responsible for the increase of energy difference between energy states and band gap. Quantum dots offer great potential in the form of QLEDs which are made out of networks of quantum dots and can also build on, yet dramatically improve, existing LED technologies.

Quantum dots are essentially nanometer-size crystals of semiconductor materials (e.g., silicon or germanium) for which the electronic properties are strongly dependent on their size.

Wavelengths in quantum dots can be controlled in nanocrystalline materials. The energy separation between valence and conduction bands can be altered in nanocrystalline quantum dots by changing the size of the nanoparticles. Resulting energy levels can thus be varied.

One-dimensional nanomaterials: The diameters of the nanostructures are comparable to the sizes of biological and chemical species being sensed, and thus intuitively represent excellent primary transducers for producing signals that ultimately interface with macroscopic instruments.

The size-tunable colours of semiconductor nanocrystals, together with their highly robust emission properties, are opening up opportunities for labelling and optical-based detection of biological species that offer advantages compared with conventional organic molecular dyes widely used today.

The underlying mechanism for nanowire sensors is a field effect that is transduced using field effect transistors (FETs), the ubiquitous switches of the microelectronics industry. The electronic characteristics of nanowires are well controlled during growth in contrast to carbon nanotubes. Si nanowires can be prepared as single-crystal structures with diameters as small as 2-3 nm.

The Einstein relation for the diffusivity (D) to the mobility (μ) ratio (DMR) of the carriers in electron devices is very useful since the diffusion constant can be obtained from this ratio by knowing the experimental values of the mobility. It is more accurate than any of the individual relations for the diffusivity or the mobility, which is the two widely, used quantities of carrier transport of modern semiconductor devices. The classical value of the DMR is equal to $(k_B T / e)$, (k_B , T , and e are Boltzmann constant, temperature and the magnitude of the carrier charge respectively) which represents the well-known Einstein relation. The relation is valid for both the electrons and the holes. In this conventional form it appears that, the DMR increases linearly with T and is independent of carrier concentration. This relation holds only under the condition of carrier non-degeneracy although its validity has been suggested erroneously for degenerate materials. Besides, the performance of the electron devices at the device terminals and the speed of operation of modern switching transistors are significantly influence by the degree of carrier degeneracy present in these devices. The simplest way of analyzing such devices taking into account of the degeneracy of the band is to use the appropriate Einstein relation to express the performance at the device terminal and switching speed in terms of the carrier concentration.

The photo-current density depends on the density-of-states function which significantly affects the different physical properties of electronic materials having various band structures. Photoemission from electronic materials having degenerate electron concentration is essentially determined by their respective energy band structures. It has different values in different materials and varies with the electron concentration, with the magnitude of the reciprocal quantizing magnetic field under magnetic quantization, with the quantizing electric field as in inversion layers, with the nano-thickness as in quantum wells, wires and dots and also with superlattice period as in the quantum confined superlattices of small gap compounds with graded interfaces having various carrier energy spectra.

This method can predict the confined energy levels of nanostructures by solving Schrodinger equation assuming the barriers have an infinite confining potential. The “effective mass” solutions of the Schrödinger equation for electrons confined in a quantum dot or NCs, quantum wire and quantum well are,

Quantum dot or Nanocrystals:

$$E_{n,m,l} = \frac{\pi^2 \hbar^2}{2m^*} \left(\frac{n^2}{L_z^2} + \frac{m^2}{L_y^2} + \frac{l^2}{L_x^2} \right), \psi = \phi(z)\phi(y)\phi(x)$$

Quantum wire:

$$E_{n,m}(k_x) = \frac{\pi^2 \hbar^2}{2m^*} \left(\frac{n^2}{L_z^2} + \frac{m^2}{L_y^2} \right) + \frac{\hbar^2 k_x^2}{2m^*}, \psi = \phi(z)\phi(y)\exp(ik_x x)$$

Quantum well:

$$E_n(k_x, k_y) = \frac{\pi^2 \hbar^2 n^2}{2m^* L_z^2} + \frac{\hbar^2}{2m^*} (k_x^2 + k_y^2), \psi = \phi(z)\exp(ik_x x + ik_y y)$$

where $n, m, l = 1, 2, \dots$ the quantum confinement numbers, L_x, L_y and L_z are the confining dimensions, $(\exp ik_x x + ik_y y)$ is the wave function describing the electronic motion in x and y direction, similar to free electron wave functions

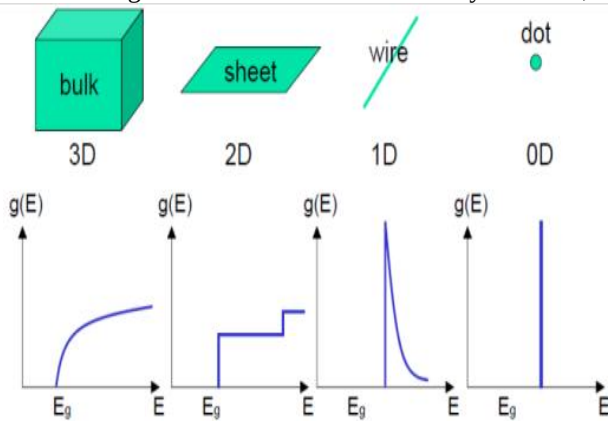


Fig. 3.2 Density of electron states of a semiconductor as a function of dimension.

The optical absorption spectrum is roughly proportional to the density of states.

The in-plane effective electron mass (m_i) in narrow Ga_{0.47}In_{0.53}As/InP quantum wells is strongly dependent on the quantization energy.

III-V materials are being increasingly used in integrated optoelectronics, passive filter devices, distributed feedback lasers and Bragg reflectors.

In recent years, with the advent of MBE, MOCVD and other experimental techniques, the influence of quantization of band states on the different physical properties of nanostructure materials such as quantum wells, quantum well wires, quantum dots, inversion layers, magnetic quantization, and different field added dimensionally reduced systems. The influence of band structures on the physical properties of quantized structures is becoming increasingly important.

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Quantum Confinement

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Abstract

Quantum Confinement is the spatial confinement of electron-hole pairs (excitons) in one or more dimensions within a material and also electronic energy levels are discrete. It is due to the confinement of the electronic wave function to the physical dimensions of the particles. Furthermore this can be divided in to three ways: 1D confinement (free carrier in a plane) -Quantum Wells, 2D confinement (carriers are free to move down) - Quantum Wire and 3D-confinement (carriers are confined in all directions) are discussed in details. In addition the formation mechanism of exciton, quantum confinement behaviour of strong, moderate and week confinement have been discussed below.

Keywords: Quantum dots; Energy level; Exciton; Confinement; Bohr radius;

1. Introduction of Quantum confinement

The term 'quantum confinement' is mainly deals with energy of confined electrons (electrons or electron-hole). The energy levels of electrons will not remain continuous as in the case of bulk materials compare to the nanocrystal. Moreover, obtains the confined electron wave functions, they become a discrete set of energy levels as shown in the figure 1. Such kinds of effects appear when the dimensions of the potential approach nearly to de Broglie wavelength of electrons and resulting in the changes or discreteing levels of energy. The effects are defined as quantum confinement and consequently, for nano-crystals are often called quantum dots (QDs). Furthermore, this quantum dot effect influence into the nanomaterial properties such as electrical, optical as well as mechanical behavior of material . It is due to the peculiar nature in nano material possess higher energy electrons than the bulk material's. Depending on the QDs size, confined electrons have higher energy than the electrons in bulk materials.

The semiconductor nanomaterials exhibit fascinating properties when reducing their dimensionality from 2D to 1D or 1D to 0D. Perhaps, the quantum confinement effect occurs when reducing the size and shape of nanomaterials is less than 100-10nm or even lesser. These changes due to discrete set of electron energy level and that lead to size confinement [1-3] figure 1&2.

To know more about quantum confinement it is necessary to understand the phenomenon of quantum dots (QDs). QDs are the new class of materials in which quantum confinement effects can be evident. QDs are very tiny semiconductor crystals in the order of nanometer size and also molecules are tightly confined electrons or electron-hole pairs called “excitons” (explained in the next section) in all three dimensions. QDs are a subatomic group in the family of nanomaterials, which comprises metals, insulators, semiconductors, and organic materials. It's well known, the quantum confinement occurs only in semiconductors quantum dots and because of their tunable bandgap nature than zero band gap in metals respectively. As mentioned before the peculiar tunable band gap properties QD are composed only group such as II-VI, III-V, and IV-VI based materials. The optical, electrical and bandgap are tunable with respect to changes in particle size that leads to different multiple application.

The main discussion is that, how bandgap can be tuned with respect to size? For that we need to understand the formation of discrete energy levels and the formation of excitons.

2. Formation of discrete energy level

To understand or recall the formation of discrete energy level, when atoms are brought together in a bulk material the number of energy states increases substantially to form nearly continuous bands of states. And also decreasing trend was occurred in the amount of atoms in material and energy states were delocalized with confinement nature. Furthermore, electron-hole pairs converting spatially confined nature when the particle towards natural de-Broglie wavelength of electrons in the conduction band. As a result the energy difference between energy bands is increased with decreasing in particle size dimension as shown in the figure 3.

Particle behaves like a free particle when the dimensions of the confining structure are very large in comparison to the de Broglie wavelength. On this stage, the energy states are continuous and the bandgap is comes to its original position and another energy spectrum does not remain continuous and becomes discrete in nature when the dimensions of confining structure are decreased towards nanoscale. Therefore, the bandgap exhibits size dependence properties, and eventually causes a blue shift in the emitted light as the particle's size is decreased. However, this effect demonstrates the consequences of confining the electrons and electron-hole pair (or the excitons) within a dimension which approaches the critical quantum limit, often termed as the Bohr exciton radius.

In this view, a quantum dot confines in all the three dimensions; a quantum wire (nanowire) confines in two dimensions; and a quantum well confines only in one dimension. The corresponding structures are also termed as zero dimensional (0D), one dimensional (1D), and two dimensional (2D) potential wells, respectively, with regard to the number of dimensions in which the confined particle has freedom of movement. The below figure 4 shows the overview of quantum confinement in nanostructures.

- Electrons confined in one direction: i.e **Quantum Wells** (thin films): Electron can easily move in 2 dimensions (**2D**) so, one dimensional is quantized.
- Electrons confined in two directions: i.e **Quantum Wires**: Electrons can easily move in 1 dimension (**1D**) so, two dimensional is quantized.
- Electrons confined in three directions: i.e. **Quantum Dots**: electrons can easily move in zero dimensions (**0D**), hence three dimensional is quantized.

In conclusion, each confinement direction changes a continuous k component to a discrete component characterized by a quantum number n .

3. Formation of Excitons

It is very important and necessary to understand the concept of excitons as it is the primary steps to understand QDs and as well as quantum confinement in semiconductors. In the case of semiconductor, electron where shifted from valence band to conduction band when emitting light falling on it and consequently recombination effect impose or create the photon particle. The electron and hole were occupied or created from conduction band and valence band respectively. However, charge of hole is equivalent to the electron charge which is helping to implementation of one particle named as excitation. Above mentioned charges and Coulomb exchange interaction there is an attractive connection between the electron and the hole. Such kind of electron-hole pair is some time expressed in simple term quasiparticle which is named exciton. It is an electrically neutral quasiparticle that occurs in insulators, semiconductors and some liquids.

In solid state physics, the band gap/energy band gap is separated between the finite energy level of conduction and valence band. When an electron from the valence band attains sufficient energy to overcome the energy gap, due to thermal excitation or absorption of a photon, and its goes to the conduction band, a hole is created on the left behind on the valence band. The created hole is moving to conduction band this is formed as an excited electron, the charge carrier in semiconductor device or recombines with hole after time the release of energy. The combination between the electron and hole pair combination leads to the formation of excitons.

Because of the different polarity charges and the Coulomb force exchange interaction there is an attractive connection between the electron and the hole and by simple way the electron-hole pair is were called as a quasiparticle which is named exciton. Due to combination of electron-hole resulting neutral quasiparticle nature were exists into different material nature like semiconductor, insulator and some of the liquid. Furthermore this exciton is transport the energy without compromising net electric charge as per condensed matter theory.

However, there is a major difference. Excitons have an average physical separation between the electron and hole, referred to as the Exciton Bohr Radius this physical distance is different for each material. In the case of semiconductor nanocrystal (QDs) the size of the particles is lesser than the Bohr radius, the electron excited by an external energy source tends to form the weak bond with its hole. This bound state of electron and electron hole, which are attracted by the electrostatic Coulomb force if often called an exciton which was shown in figure 5A&B.

Thus, the Bohr radius is the distance in an electron-hole exciton, also called the exciton Bohr radius [4]. Every semiconductor material has a characteristic exciton Bohr radius (Fig. 5C&D) below which the quantum confinement effect is realized. This unique

confinement property causes the “band” of energies to turn into discrete energy levels in QDs (fig 6.)

3.1 Excitons

To understand the quantum mechanical behaviour of semiconductor particle, let consider materials in an electron which is stimulated from valance band to conduction band (figure 7).

From the valance band left behind electron is a ‘hole’, which can be thought of as a particle with its own charge (+1) and effective mass. The bound state of electron and hole are considered to be quasiparticles is then known or called as ‘exciton’. The pair of electron and hole formation due to Columbic attraction. The exciton can be considered as hydrogen like system, and the exciton radius is easily compare with Bohr radius approximation which is used to calculate the spatial separation of the electron-hole pair,

$$r = \frac{\epsilon \hbar^2}{\pi m_r e^2} \dots\dots\dots (1)$$

Where, r - is the radius of three dimensional sphere containing excitation, m_r - is the reduced mass of the excitation, ϵ -is the dielectric constant of the given material, and e - is the charge of the electron and $\hbar$ - is plank’s constant. Ion cyclotron resonance is a powerful technique to find the effective mass of hole and electron and are range were 0.1-3 m_e (m_e -is the mass of the electron) .Furthermore, the empirical radius and mass calculation suggests that the electron-hole pair spatial separation is around about 1-10 nm for most the semiconductors.

4. Quantum confinement effect:

In literature, semiconductor quantum dots are also known as semiconductor nanocrystals or nanoparticles. A semiconductor nanocrystal (SNC) or quantum dot (QD) is a semiconductor whose excitons are confined in all three spatial dimensions. As a result, they have properties that are between bulk semiconductors and those of discrete molecules. Size effects are observed in semiconductor crystals measuring 10-100 nm, whereas quantum size effects are usually the characteristics of nanocrystallites measuring less than 10 nm. The physico chemical properties of nanocrystalline particles different from those of bulk materials for two specific reasons. First, the high surface to volume ratio that results in many atoms at the surface in the crystalline lattice. And secondly, the electronic bands are spilt up in to discrete energy levels as the result three dimensional confinement of the charge carriers occurs. This leads to quantum confinement of charge results in the increase bandgap with decreasing particles size.

The quantum confinement effect in low dimensional semiconductor systems was described about 25 years ago. The bulk crystalline structure is preserved in a nanocrystal. However, due to quantum confinement, nanocrystals have molecule-like discrete electronic states which exhibit strong in size dependent properties. In the last decade, comprehensive investigations were made to explore size dependent properties of semiconductors with emphasis on optical properties, including absorption and luminescence.

Detail knowledge of the optical properties of the nanostructured materials is important for understanding the photophysical and photochemical process that follow the absorption of light quanta. Steigewald and Brus (1990)[5] used quantum mechanical molecular orbital calculations to explain quantum confinement effect on optical absorption, accounting the well established blue shift in the absorption spectra with decreasing particle size. Bawendi (1995) [6-7] has also provided a recent overview of this field. As explaining observed spectral shifts during the early stages of inorganic semiconductor particle growth, molecular orbital (MO), and linear combination of atomic orbital coupled with molecular orbital (LCAO-MO) procedures provide information for the construction of energy level diagrams for clusters of several molecules up to size characteristic of bulk semiconductor.

By addition of the filled and empty orbitals of the multiple energy levels can increase the molecules level in a cluster. This decreasing trend in energy differences between the not only filled orbitals and as well as empty orbitals too. Also, decreasing trend into the energy gap was observed between the highest occupied molecular orbitals (HOMO) and the lowest unoccupied molecular orbitals (LUMO). For a bulk semiconductor, the filled and empty state form separated continuums i.e., the valence and conduction bands. However, for Quantum Dot (Q-size) regime region energy levels were within the empty and filled states remain discrete and the band gap energy levels were higher states of between the HOMO and the LUMO than of the bulk materials as shown in the schematic diagram (Figure 8).

The band gap, E_g , increases in magnitude as the semiconductor particle radius decrease in size to the point where it becomes comparable or smaller than the exciton radius. These properties have led to the applications in a wide range of fields i.e., semiconducting quantum well and super lattice devices, nonlinear optical materials, photocatalysis and imaging systems. For II-VI compounds, maximum radii for the onset of quantum confinement effect (Q-size effects) leading to the bandgap increase have been determined by Bawendi (1995)[7] to be 10 to 100 Å. Wide band photoconductors such as, TiO_2 (rutile) and ZnO undergo increase in band gap as radii approach in the range of 50 Å. It is interesting to note that a blue shift in the long wavelength optical absorption edge with decreases in particle size was first observed by Berry (1967)[8] for AgBr samples with radii of 65 Å and 350 Å. It's very clear that Quantum confinement effect based semiconductor nanocrystals were produced tremendous optical and electrical properties than of the bulk materials which are due to the reduced behavior of characteristic length which is called as exciton Bohr diameter. This peculiar characteristic length is usually in the range of few nanometers when the material reveals size dependent of optical and electrical properties.

One of the most important consequences of the spatial confinement effect is an increase in energy of the band-to-band excitation peaks (blue shift) as the radius R of a microcrystallite semiconductor is reduced in relation with the Bohr radius. However due to significant spatial confinement effect there is an increase in energy of band-to-band excitation peaks which is called blue shift. In terms, microcrystallite of Radius (R) ranges in semiconductor having less relation with than of the Bohr Radius. Some of minor difference would have happen between the theoretical and experimental of confine effect it is due to electron electron-hole interaction energy, in coulomb term and the confinement energy of the electron and hole in the kinetic energy.

Quantum confinement

- When the length of a semiconductor is reduced to the same order as the exciton radius, i.e., to a few nanometers, quantum confinement effect occurs and the exciton properties are modified. Depending on the dimension of the confinement, three kinds of confined structures are defined: quantum well (QW), quantum wire (QR) and quantum dot (QD)
- ✓ The reduction in dimensionality produced by confining electrons (or holes) to a thin semiconductor layer leads to a dramatic change in their behavior.
 - ✓ This principle can be developed by further reducing the **dimensionality** of the electron's environment from a two-dimensional quantum well to a one -dimensional quantum wire and eventually to a zero-dimensional quantum dot.
 - ✓ The dimensionality refers to the number of degrees of freedom in the electron momentum ; in fact, within a quantum wire , the electron is confined across two directions, rather than just the one in a quantum well and so therefore reducing the degrees of freedom to one.
 - ✓ In a quantum dot, the electron is confined in all three dimensions, thus reducing the degrees of freedom to zero.
 - ✓ If the number of **degrees of freedom** are labeled as D_f and the number of directions of confinement are labeled as D_c then clearly:

$$D_f + D_c = 3$$

Structure	Degree of Confinement (D_c)	Degree of freedom (D_f)
Bulk Material	0D	3D
Quantum Well	1D	2D
Quantum Wire	2D	1D
Quantum Dot	3D	0D

Exciton Bohr Radius and Quantum Confinement

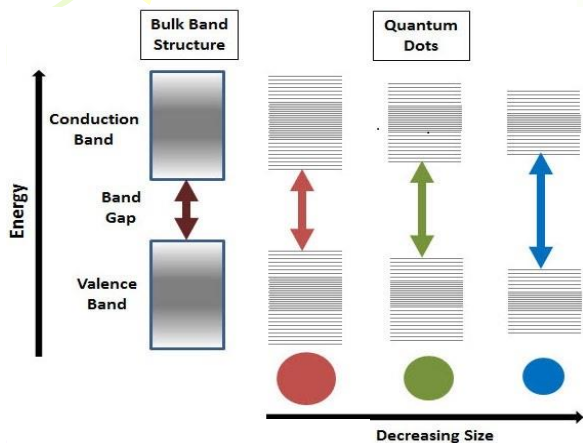
Bohr Radius (1913):

The most probable distance between the proton and electron in a hydrogen atom in its

Excitons are coupled electron-hole pairs via Coulomb attraction

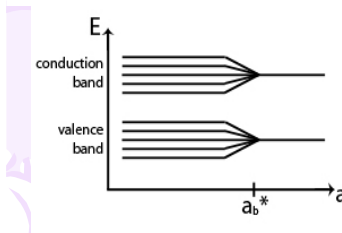
$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} = 0.529 \text{ \AA}$$

$$a_{\text{exciton}} = \frac{a_0 \epsilon}{m^* / m_e}$$



➤ When the length of a semiconductor is reduced to the same order as the exciton radius, i.e., to a few nanometers, quantum confinement effect occurs and the exciton properties are modified. Depending on the dimension of the confinement, three kinds of confined structures are defined: quantum well (QW), quantum wire (QR) and quantum dot (QD)

splitting of energy levels in quantum dots due to the quantum confinement effect, semiconductor band gap increases with decrease in size of the nanocrystal.



Quantum confinement of electronic particles in nanocrystals produces unique optical and electronic properties that have the potential to enhance the power conversion efficiency of solar cells for photovoltaic.

Quantum confinement of different nanostructures:

- Nanostructured materials are those with at least one dimension falling in nanometer scale, and include nanoparticles (including quantum dots, when exhibiting quantum effects), nanorodes and nanowires, thin films, and bulk materials made of nanoscale building blocks or consisting of nanoscale structures.

In general, nanomaterials have extremely small size which having at least one dimension 100 nm or less.

In terms of directions of confinements, **nanomaterials** can be nanoscale in one dimension (eg. surface films), two dimensions (eg. strands or fibres), or three dimensions (eg. particles).

In terms of degree of freedom, Siegel classified the **nanostructured** materials as Zero dimensional (quantum dot), one dimensional (quantum wire), two dimensional (quantum well), three dimensional (bulk system) nanostructures as shown in the figure 1.

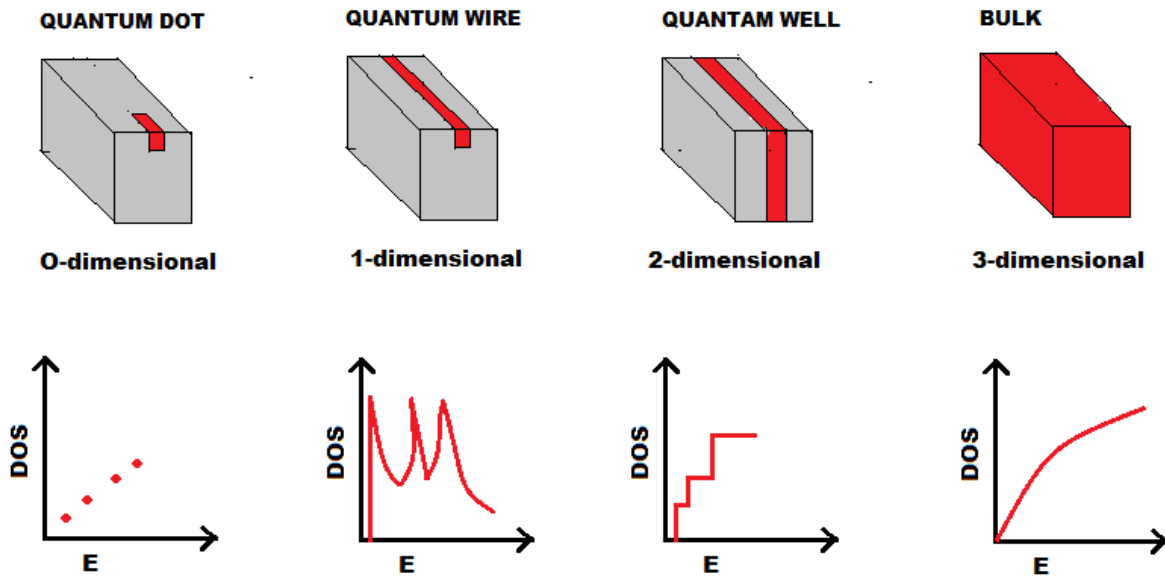


Figure 1: Quantum confinement of different nanostructures as 0-D, 1-D, 2-D and 3-D with their density of states (DOS) effects.

Zero-dimensional nanostructures:

- ✓ Zero dimensional **nanostructure** is called **quantum dot** (QD) nanocrystal. Generally, nanocrystal is a semiconductor crystal whose size is on the order of just a few nanometers.
- ✓ They contain anywhere from 100 to 1,000 electrons and range from 2 to 10 nanometers, or 10 to 50 atoms in diameter. QDs are unique because of their size and properties.
- ✓ Almost all materials system including metal, insulators and semiconductors show size dependent electronic or optical properties in the **quantum size** regime.
- ✓ Among these, the modification in the energy band gap of semiconductors is the most attractive one because of the fundamental as well as technological importance.
- ✓ Semiconductors with widely tunable energy band gap are considered to be the materials for next generation flat panel displays, photovoltaic, optoelectronic devices, laser, sensors, photonic band gap devices, etc.

One-dimensional nanostructures:

- ✓ One-dimensional nanostructures are called nanowires.

- ✓ Nanowires are attracting much interest from those seeking to apply **nanotechnology** especially in solid-state electronics and diagnosis tool in medical sciences.
- ✓ Unlike other low-dimensional systems, nanowires have two quantum-confined directions but one unconfined direction (degree of freedom) available for electrical conduction.
- ✓ This makes nanowires to be used in applications where electrical conduction, rather than tunneling transport, is required.

Two-dimensional nanostructures:

- ✓ Two-dimensional nanostructure is referred as **quantum well** in nanoscience and **nanotechnology**.
- ✓ A quantum well is a particular kind of heterostructure in which one thin "well" layer is surrounded (sandwiched) by two "barrier" layers.
- ✓ Both electrons and holes perceive lower energy in the "well" layer, hence the name (by analogy with a "potential well").
- ✓ This layer, in which both electrons and holes are confined, is so thin (typically about 100 Å, or about 40 atomic layers) that we cannot neglect the fact that the electron and hole are both waves.
- ✓ In fact, the allowed states in this structure correspond to standing waves in the direction perpendicular to the layers. Because only particular waves are standing waves, the system is quantized, hence the name "quantum well".
- ✓ Quantum wells are thin layered semiconductor structures in which we can observe and control many quantum mechanical effects.
- ✓ They can be made to a high degree of precision by modern epitaxial crystal growth techniques.